

COSMO Cube

COSMO® Cube

Hardware Reference Manual

LOT Systems, Inc. — COSMO® CIA Hardware Division

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Classification: Engineering Reference · Inventor: Vadim Marmeladov

COSMO® Cube — Device Specification v1.0

Document: COSMO-DEVICE-SPEC-v1.md

Product: COSMO® Cube

Classification: Engineering Specification — Internal

Author: Vadim Marmeladov, Inventor, COSMO® CIA

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Status: Draft for PCBWay Submission

1. Product Overview

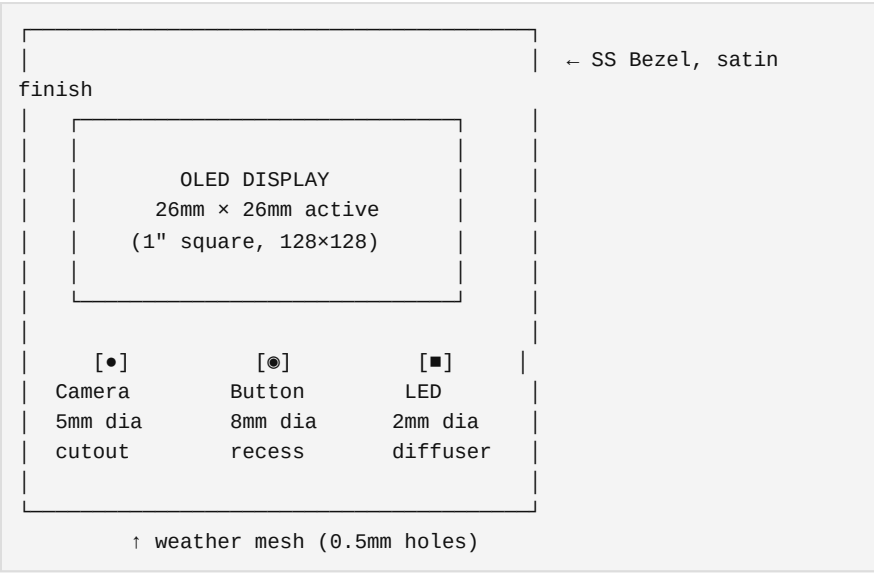
The COSMO® Cube is a personal ambient intelligence device — a hardware node connected to the LOT Systems platform at lot-systems.com. It sits on a desk or pocket, receives autonomous notifications from the AI-powered LOT site, captures environmental data, and sends behavioral signals back via a single physical button.

This is not a smartphone. It is a dedicated companion device for LOT Users with a Benchmark score of Purple or higher — an ambient window into their LOT operating system.

2. Physical Specifications

Parameter	Value
Form Factor	Flat square tablet
Dimensions	40mm × 40mm × 5mm
Weight	~28g (target, including battery)
Body Material	316L Stainless Steel, CNC Machined
Body Parts	2 (Base plate + Front bezel, snap/screw join)
Side A (Back)	Mirror-polished stainless steel — #8 finish
Side B (Front)	Satin/bead-blasted surface, functional face
Color	Natural silver stainless
Engraving	LOT® wordmark, laser engraved, Side A center
COSMO® mark	Laser engraved, Side A bottom-right, 4pt
Water Resistance	IP54 (splash resistant)
Operating Temp	0°C to +45°C
Storage Temp	-20°C to +60°C

3. Front Face Layout (Side B — 40mm × 40mm)



Front Face Elements

Element	Spec	Position
OLED Display	1.0" square, 128×128 px, SSD1327 controller	Top-center, 7mm from top edge
Camera	2MP, f/2.2, fixed focus, flush-mounted	Bottom-left, 8mm from left, 5mm from bottom
Copy Button	Stainless tactile dome, 8mm diameter	Bottom-center
Status LED	RGB LED behind frosted SS diffuser	Bottom-right
Weather Mesh	Micro-perforated SS, 5×5mm area	Right edge, center height
Charging Contact	Hidden flush pogo pins (2×) for dock alt.	Bottom edge

4. Back Face (Side A — Mirror Polished)

- Mirror polished to #8 finish (reflective, no grain)
 - LOT® wordmark engraved center (0.3mm deep laser etch, filled black)
 - COSMO® CIA engraved lower-right (0.2mm deep)
 - Serial number engraved lower-left (format: CQ-[UNIT]-[YEAR], e.g. CQ-001-26)
 - No cutouts, ports, or protrusions
 - Wireless charging coil embedded behind this surface
-

5. Electronics Architecture

5.1 System-on-Module: ESP32-S3-MINI-1U

Parameter	Value
MCU	ESP32-S3, Xtensa LX7 dual-core, 240MHz
AI Acceleration	512KB vector instructions (on-device inference)
RAM	512KB SRAM + 2MB PSRAM (external)
Flash	8MB NOR SPI Flash
WiFi	802.11b/g/n 2.4GHz, WPA3
Bluetooth	BLE 5.0
Module Size	15.4mm × 20.5mm × 3.1mm
GPIO Count	36 usable
ADC	12-bit, 20 channels
Temperature Range	-40°C to +85°C

Why ESP32-S3: Native WiFi for real-time LOT API polling, built-in vector acceleration for local inference (notification filtering), proven ecosystem, PCBWay-certified module.

5.2 Display: SSD1327 OLED 1.0” Monochrome

Parameter	Value
Controller	Solomon Systech SSD1327
Resolution	128×128 pixels
Color	16-level grayscale (4-bit)
Interface	SPI (4-wire, up to 10MHz)
Active Area	26mm × 26mm
Module Dimensions	32mm × 32mm × 1.6mm
Viewing Angle	>160°
Contrast Ratio	>2000:1
Supply Voltage	3.3V (logic), internal DC-DC for OLED
Power (active)	~20mA
Power (idle)	<1mA

5.3 Camera: Himax HM01B0

Parameter	Value
Sensor	Himax HM01B0 (ultra-low power CMOS)
Resolution	QVGA 320×320 / QCIF 176×144
Interface	DVP (8-bit parallel) or MIPI
Frame Rate	60fps (QVGA)
Power	1.1mW at 30fps QVGA
Package	CSP 2.45mm × 2.45mm × 1.6mm
Lens	M7 fixed-focus, f/2.2, HFOV 66°
AI Features	On-chip AE, AWB, histogram

Primary use: Face detection for owner recognition (COSMO® Soul Sync gate), QR code capture for pairing, activity logging image snapshot.

5.4 Weather & Environmental Sensors

Primary Weather: Bosch BME280

Parameter	Spec
Part	Bosch BME280
Measures	Temperature, Humidity, Barometric Pressure
Temperature Range	-40°C to +85°C, ±0.5°C
Humidity Range	0–100% RH, ±3% RH
Pressure Range	300–1100 hPa, ±1 hPa
Interface	I2C (up to 3.4MHz) or SPI
Package	LGA 2.5×2.5×0.93mm
Current	3.6µA at 1Hz sampling

AI-Grade IMU: TDK InvenSense ICM-42688-P

Parameter	Spec
Part	TDK ICM-42688-P
Type	6-Axis IMU (3-axis accel + 3-axis gyro)
Accel Range	±2g to ±16g, 0.004mg/LSB noise
Gyro Range	±15.625 dps to ±2000 dps
Interface	SPI (up to 24MHz) or I2C
Package	QFN 2.5×2.5×0.91mm
Current	680µA (full operation)
Use Case	Motion detection, orientation, gesture, tap-to-wake

Ambient Light + Gesture: Broadcom APDS-9960

Parameter	Spec
Part	Broadcom APDS-9960
Measures	Gesture, Proximity, ALS, RGBC color
Gesture Range	Up/Down/Left/Right, 10–20cm
Interface	I2C
Package	LCC 3.94×2.36×1.35mm

Parameter	Spec
Current	0.001μA (sleep)
Use Case	Wake on hand approach, notification dismiss gesture

5.5 Power Architecture

Battery

Parameter	Value
Chemistry	Lithium Polymer (LiPo)
Capacity	280mAh (custom cell)
Dimensions	35mm × 35mm × 2.5mm
Nominal Voltage	3.7V
Max Charge	4.2V
Discharge Cutoff	3.0V
Connector	JST 1.25mm 2-pin
Estimated Life	~18 hours active, ~4 days standby

Wireless Charging

Parameter	Value
Standard	Qi (WPC 1.3)
Max Power	5W receiver
Coil	Round, 30mm diameter, 0.4mm thick
Receiver IC	Texas Instruments BQ51013B
Charge Voltage	5V → regulated to 4.2V
Efficiency	~85% at alignment
Indicator	LED pulse during charging

Power Management IC: Texas Instruments BQ25892

Parameter	Value
Part	TI BQ25892 (or BQ25895)
Input	5V via Qi, up to 9V
Charge Current	100mA–3A programmable
Regulation	4.2V (LiPo), $\pm 1\%$
System Output	3.3V @ 500mA (for MCU, sensors)
I2C	Configuration via ESP32-S3
Protection	OVP, OCP, OTP, short circuit

Regulated Rails

Rail	Voltage	Source	Consumers
VSYS	3.7V (battery)	LiPo	BQ25892 input
VCC	3.3V	BQ25892 SYS	ESP32-S3, sensors
VOLED	7-9V	OLED internal boost	Display
VCAM	1.8V	LDO (AP2112K)	Camera

6. Connectivity

Protocol	Use
WiFi 802.11n 2.4GHz	LOT API polling, OTA firmware updates
BLE 5.0	Companion app pairing, local config
I2C Bus (400kHz)	BME280, APDS-9960, OLED
SPI Bus (10MHz)	ICM-42688-P, additional flash
DVP / MIPI	Camera
2× Pogo Pins (bottom edge)	Optional dock charging contact + UART debug

7. LOT Platform Integration

7.1 Copy Button Function

Pressing the physical Copy button: 1. Captures current environmental snapshot (temp, humidity, pressure, light, orientation) 2. Captures camera frame (optional, user-configurable) 3. Packages as JSON payload 4. POSTs to `https://lot-systems.com/api/hardware/log` 5. Includes device serial, timestamp (UTC), sensor data, user token 6. Site displays entry in **Log tab** with tag `[COSMO® cube]` 7. Device LED pulses green (success) or red (failed, retries 3×)

7.2 Notification Display

- Device polls `https://lot-systems.com/api/hardware/notifications` every 60 seconds via WiFi
- Alternatively: maintains persistent WebSocket/SSE connection to `lot-systems.com`
- Received notification JSON → rendered on OLED display
- Display format: notification text (e.g., “Coffee time!”), source, timestamp
- Max display duration: 30 seconds, then returns to clock/status screen
- Notification queue: up to 10 pending, FIFO

7.3 Authentication

- Device provisioned with unique API key at manufacture time
- API key stored in ESP32-S3 NVS (encrypted flash partition)
- TLS 1.3 for all API calls, certificate pinned to `lot-systems.com` CA
- Key rotation: OTA update via secure firmware channel

8. Display Content States

State	Screen Content
Clock (default)	Time HH:MM, date, weather temp
Notification	Message text, LOT logo small, source label

State	Screen Content
Charging	Battery percentage + Qi indicator animation
Log Sent	“Logged.” confirmation, 3 seconds
Pairing	QR code for app pairing
Sleep	Display off (ambient light < threshold)
Low Battery	Battery icon, percentage, charge reminder
Error	Error code + short descriptor

9. Firmware Summary

See COSMO-FIRMWARE-v1.md for full specification.

Layer	Technology
RTOS	FreeRTOS (via ESP-IDF 5.2)
Language	C / C++17
WiFi Stack	ESP-IDF lwIP + mbedTLS
Display Driver	Custom SSD1327 SPI driver
Camera Driver	ESP32-S3 Camera (esp32-camera component)
Sensor Drivers	I2C/SPI HAL layer
OTA	ESP-IDF OTA (HTTPS signed)
Secure Boot	RSA-3072 signed bootchain
Flash Encryption	AES-256 XTS

10. Regulatory Targets

Certification	Region	Status
FCC Part 15	USA	Required before US sale
CE Mark	EU	Required before EU sale
IC	Canada	Required before Canada sale

Certification	Region	Status
RoHS	Global	PCBWay ensures compliance
REACH	EU	Material compliance via SS 316L
Qi Certification	Global	WPC certification required

Note for 100-unit production run: Pre-certification development units only. FCC/CE required before commercial sale. Budget ~\$15,000–\$25,000 for full certification.

11. Power Budget

Condition	Current Draw	Battery Life
Active WiFi, display on	~180mA	~1.5 hours
WiFi polling (60s interval), display on	~45mA	~6 hours
Deep sleep, wakeup every 60s	~2mA	~5.8 days
Charging (Qi, 5W)	N/A	~2.5 hours full charge

Target real-world: 18 hours with WiFi polling every 60s, display active 20% of time.

Pin Assignments (ESP32-S3-MINI-1U)

```
#define DISPLAY_MOSI      GPIO_NUM_11
#define DISPLAY_CLK       GPIO_NUM_12
#define DISPLAY_CS        GPIO_NUM_10
#define DISPLAY_DC        GPIO_NUM_9
#define DISPLAY_RST       GPIO_NUM_8

// --- Camera (DVP parallel) ---
#define CAM_D0             GPIO_NUM_14
#define CAM_D1             GPIO_NUM_13
#define CAM_D2             GPIO_NUM_21
#define CAM_D3             GPIO_NUM_47
#define CAM_D4             GPIO_NUM_48
#define CAM_D5             GPIO_NUM_45
#define CAM_D6             GPIO_NUM_38
#define CAM_D7             GPIO_NUM_39
#define CAM_VSYNC         GPIO_NUM_6
#define CAM_HREF           GPIO_NUM_7
#define CAM_PCLK           GPIO_NUM_15
#define CAM_XCLK           GPIO_NUM_16

// --- I2C Bus (BME280, APDS-9960, BQ25892) ---
#define I2C_SDA            GPIO_NUM_1
#define I2C_SCL            GPIO_NUM_2
#define I2C_FREQ_HZ        400000

// --- SPI Bus (ICM-42688) ---
#define IMU_MOSI           GPIO_NUM_35
#define IMU_MISO           GPIO_NUM_36
#define IMU_CLK            GPIO_NUM_37
#define IMU_CS             GPIO_NUM_34

// --- Button (Copy) ---
#define BTN_COPY           GPIO_NUM_0           // GPIO0, internal
pull-up
#define BTN_DEBOUNCE_MS    50

// --- LED (RGB) ---
#define LED_R              GPIO_NUM_3
#define LED_G              GPIO_NUM_4
#define LED_B              GPIO_NUM_5
```

```
    // --- Power ---  
    #define CHG_INT          GPIO_NUM_17      // BQ25892  
interrupt pin  
    #define VSYS_EN          GPIO_NUM_18      // System power  
enable
```